CSA4707 – Deep Learning

Experiment 2: Two-Class Confusion Matrix

Aim:

To demonstrate a two-class confusion matrix using Python.

Program:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score
from sklearn.metrics import f1_score, classification_report, confusion_matrix

wine = load_wine()
data = pd.DataFrame(data=wine.data, columns=wine.feature_names)
data['Target'] = wine.target

data = data[data['Target'] != 2]

X = data.drop('Target', axis=1)
y = data['Target']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)

model = DecisionTreeClassifier(random_state=1)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

class_report = classification_report(
    y_test, y_pred, target_names=wine.target_names[:2]
)
print("Classification Report:\n", class_report)
```

```
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print(f"Precision: {precision:.2f}")
print(f"Recall: {recall:.2f}")
print(f"F1 Score: {f1:.2f}")

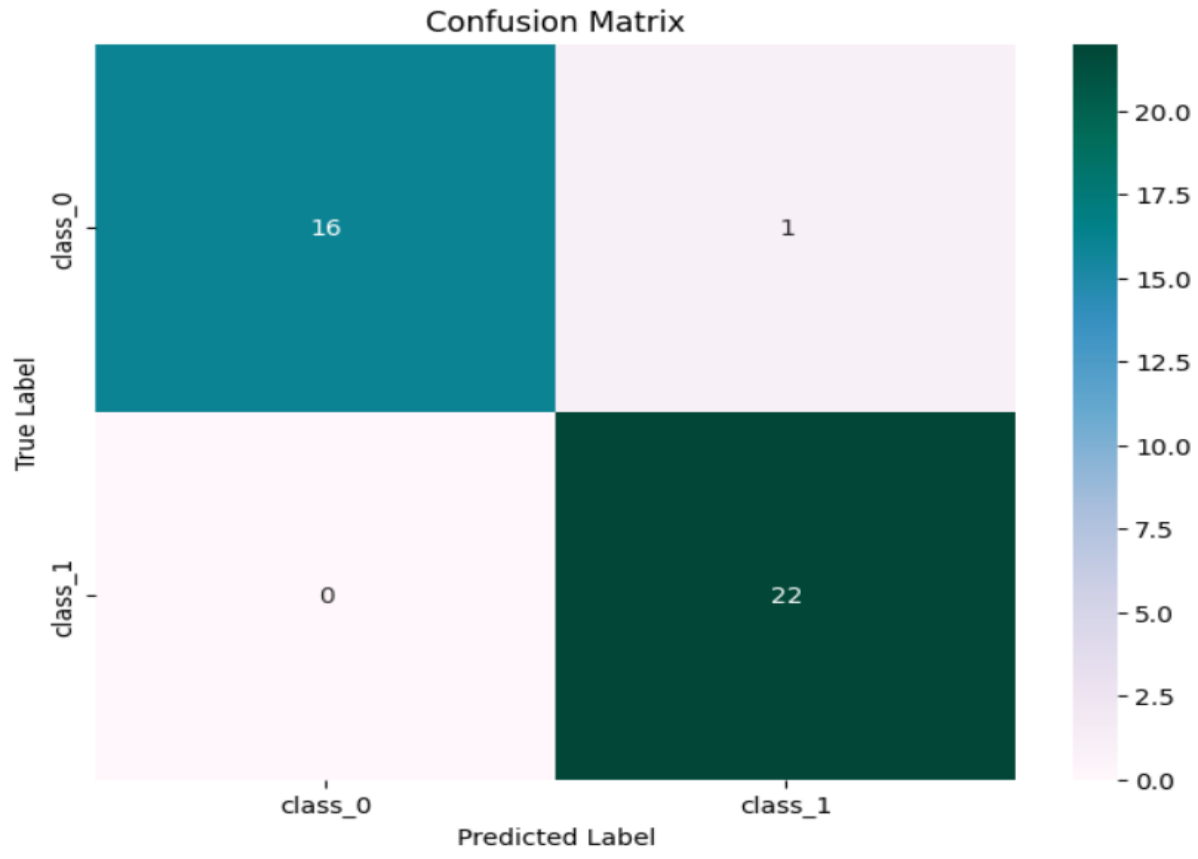
conf_matrix = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(8, 6))
sns.heatmap(
    conf_matrix, annot=True, fmt='d', cmap='PuBuGn',
    xticklabels=wine.target_names[:2],
    yticklabels=wine.target_names[:2]
)
plt.xlabel("Predicted Label")
plt.ylabel("True Label")

plt.title("Confusion Matrix")

plt.show()
```

Output:



Result / Output:

Accuracy: 0.9743589743589743

Precision: 0.96

Recall: 1.00

F1 Score: 0.98

The classification report and confusion matrix are displayed. The decision tree classifier achieves approximately 97.44% accuracy on the test data.